

Markus Tatzgern

Curriculum Vitae



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🌐 Web markustatzgern.com

🔗 Scholar scholar.google.com/citations?user=Fs0fyasAAAAJ

🆔 ORCID [0000-0002-3900-4944](https://orcid.org/0000-0002-3900-4944)

Profile

My research develops visualization and interaction techniques for Augmented and Virtual Reality that bridge the digital and the physical world. I explore adaptive visualization, multimodal interaction and AI-driven approaches for human-centered XR, with applications in training and learning, healthcare, human-robot interaction and everyday augmentation.

Publications 7 journal and 26 conference papers; 18 at CORE A* venues (e.g. ACM CHI, ACM UIST, IEEE ISMAR, IEEE VR), 6 in SJR Q1 journals; 4 best paper honorable mentions

Funding More than €3.4m acquired for my lab since 2015; involved in proposals totalling about €8.7m

Leadership Science & Technology Program Chair IEEE ISMAR 2024, Papers Chair IEEE ISMAR 2025; Associate Editor IEEE TVCG; Head of Research, Salzburg University of Applied Sciences (2023–2026)

Employment

since 09/2026 **Associate Professor of Creative Technologies**, Warwick Manufacturing Group & School of Cross-Faculty Studies, University of Warwick, England

01/2023–08/2026 **Head of Research**, Department of Creative Technologies, Salzburg University of Applied Sciences, Austria

06/2018–08/2026 **FH-Professor**, Mixed Reality and Game Development, Salzburg University of Applied Sciences, Austria

09/2015–08/2026 **Senior Lecturer, Head of Academic Area**, Game & Mixed Reality, Salzburg University of Applied Sciences, Austria

09/2009–08/2015 **Research Assistant**, Christian Doppler Laboratory for Handheld Augmented Reality, Institute of Computer Graphics and Vision, Graz University of Technology, Austria

Education

2015 **Dr.techn. in Computer Science** (Doktor der technischen Wissenschaften, equivalent to PhD)

Graz University of Technology, Institute of Computer Graphics and Vision, Austria

Thesis: *Situated Visualization in Augmented Reality*

Graduated with highest distinction

- 2008 **Dipl.-Ing. in Telematics** (Diplom-Ingenieur, equivalent to MSc)
Graz University of Technology, Institute of Computer Graphics and Vision, Austria
Thesis: *A Framework for Automatically Creating 3D Explosion Diagrams*
Graduated with highest distinction
- 2005 **Bakk.techn. in Telematics** (Bakkalaureus der technischen Wissenschaften, equivalent to BSc)
Graz University of Technology, Austria

Research Funding

In total, I have acquired more than € 3.4m for my lab since joining the Salzburg University of Applied Sciences in 2015 and have participated in project proposals that resulted in about € 8.7m of funding. In the following, the total project budget and the university share are given as 'total (share)'. Contract research for industry partners is not listed due to data protection and non-disclosure agreements.

Salzburg University of Applied Sciences

- 05/2025–04/2029 **GenAledTech - Content Creation with Generative Artificial Intelligence for Interactive Educational Technologies**
Funded by the **FFG** Austrian Research Promotion Agency / COIN Aufbau
Budget: 720.000€
Role: Principal investigator, project acquisition
- 04/2025–06/2027 **Extended Reality and Micro Learning for Vocational Training.**
Funded by the **Republic of Austria**, Federal Ministry Agriculture and Forestry, Climate and Environmental Protection, Regions and Water Management,
Partners: Adult Education Center Burgenland, University of Salzburg
Budget: 262.000€ (75.000€)
Role: Co-investigator, project acquisition
- 01/2025–12/2026 **TGuard: Combating Disinformation through the Analysis and Detection of Fake News Networks**
Funded by the **FFG** Austrian Research Promotion Agency / K-PASS 2023
Partner: Austrian Institute of Technology
Budget: 735.000€ (314.000€)
Role: Project development and acquisition as Head of Research
- 01/2024–12/2026 **FAIR AI: Fostering Austria's Innovative Strength and Research excellence in Artificial Intelligence**
Funded by the **FFG** Austrian Research Promotion Agency / Artificial Intelligence Mission Austria (AIM AT)
Lead partner: Austrian Institute of Technology
Budget: 3.0M€ (100.000€)
Role: Co-investigator, project acquisition
- 05/2022–04/2025 **CIRCUIT: Towards Comprehensive CBCT Imaging Pipelines for Real-time Acquisition, Analysis, Interaction and Visualization**
Funded by the Austrian Research Promotion Agency / **FFG** Bridge (no FO999891105)
Industry partner: medPhoton GmbH
Budget: 470.000€
Role: Principal investigator, project acquisition

- 05/2022–04/2026 **doc.hci – Designing Meaningful Human-Technology Relations**
 Funded by **FWF** Austrian Science Fund, doc.funds.connect
 Research partner: University of Salzburg
 Budget: 1.0M€ (417.000€)
 Role: Co-investigator, project acquisition
- 03/2021–02/2024 **EdTech Salzburg - Augmented Learning Lab**
 Funded by Federal State Government of Salzburg, Austria
 Research partner: Salzburg University of Education Stefan Zweig, University of Salzburg
 Budget: 1.1M€ (393.000€)
 Role: Principal investigator, project acquisition
- 06/2020–11/2023 **Adaptive Visualization for Augmented Reality**
 Funded by the Austrian Research Promotion Agency / **FFG** Bridge (no 877104), Austria
 Industry partner: Wikitude GmbH / Qualcomm Ltd.
 Budget: 383.000€
 Role: Project acquisition, principal investigator
- 03/2020–08/2022 **EdTech in the University Education of Healthcare Professionals**
 Funded by the Federal State Government of Salzburg, Austria
 Research partner: Health Sciences / Physiotherapy of Salzburg University of Applied Sciences
 Budget: 110.000€
 Role: Project acquisition, principal investigator
- 03/2020–08/2021 **Investigating Virtual Reality Environments for the Chemistry Curriculum**
 Funded by the Joachim Hertz Foundation, Germany
 Research partner: Paris Lodron University of Salzburg
 Budget: 18.000€ (11.000€)
 Role: Project acquisition, co-investigator
- 12/2017–11/2020 **Ecomedicine VR Physiology Lab: Information and communication technology drives innovation for health products and services**
 Funded by Federal State Government of Salzburg, Austria
 Research partner: Paracelsus Medical Private University
 Budget: 883.000€ (317.000€)
 Role: Project acquisition, co-investigator
- 03/2017–02/2019 **Optimizing the Online Shopping Experience using Virtual Reality**
 Funded by Federal State Government of State Salzburg, Austria
 Budget: 120.000€
 Role: Project acquisition, co-investigator
- [Graz University of Technology](#)
- 03/2010–08/2015 **Christian Doppler Laboratory for Handheld Augmented Reality**
 Funded by Christian Doppler Society
 Industry partner: Qualcomm Inc.
 Role: Early stage researcher
- 09/2009–02/2010 **IPCity, Integrated Project on Interaction and Presence in Urban Environments**
 Funded by European Union under contract FP-2004-IST-4-27571
 Role: Early stage researcher

Publications

Co-supervised students are marked: ° PhD student, † Master's student. Titles and DOIs link to the publication.

Journal

- 1 Gerlinde Emsenhuber°, Tobias Langlotz, Denis Kalkofen, and **Markus Tatzgern**. “See what I Mean? Mobile Eye-Perspective Rendering for Optical See-Through Head-Mounted Displays”. In: *IEEE Transactions on Visualization and Computer Graphics (TVCG), Special Issue IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* 31.11 (2025), pp. 9603–9613. DOI: 10.1109/TVCG.2025.3616739
SJR Q1, CORE A*
- 2 Arno Kinzinger, Winfried J. Steiner, **Markus Tatzgern**, and Christine Vallaster. “Interactive product presentation in an immersive environment: The influence of functional control on hedonic aspects”. In: *Journal of Retailing and Consumer Services* 83 (2025), p. 104156. DOI: 10.1016/j.jretconser.2024.104156
SJR Q1
- 3 Arno Kinzinger, Winfried Steiner, **Markus Tatzgern**, and Christine Vallaster. “Comparing low sensory enabling (LSE) and high sensory enabling (HSE) virtual product presentation modes in e-commerce”. In: *Information Systems Journal* 32.5 (2022), pp. 1034–1063. DOI: 10.1111/isj.12382
SJR Q1
- 4 Maria CR Harrington, **Markus Tatzgern**, Tom Langer†, and John W Wenzel. “Augmented reality brings the real world into natural history dioramas with data visualizations and bioacoustics at the Carnegie Museum of Natural History”. In: *Curator: The Museum Journal* 62.2 (2019), pp. 177–193. DOI: 10.1111/cura.12308
SJR Q1
- 5 Jacob Boesen Madsen, **Markus Tatzgern**, Claus B. Madsen, Dieter Schmalstieg, and Denis Kalkofen. “Temporal Coherence Strategies for Augmented Reality Labeling”. In: *IEEE Transactions on Visualization and Computer Graphics (TVCG)* 22.4 (2016), pp. 1415–1423. DOI: 10.1109/TVCG.2016.2518318
SJR Q1, CORE A*
- 6 **Markus Tatzgern**, Raphael Grasset, Eduardo Veas, Denis Kalkofen, Hartmut Seichter, and Dieter Schmalstieg. “Exploring Real World Points of Interest: Design and Evaluation of Object-Centric Exploration Techniques for Augmented Reality”. In: *Pervasive and Mobile Computing* 18 (2015), pp. 55–70. DOI: 10.1016/j.pmcj.2014.08.010
SJR Q1
- 7 **Markus Tatzgern**, Denis Kalkofen, and Dieter Schmalstieg. “Multi-perspective compact explosion diagrams”. In: *Computers & Graphics* 35.1 (2011), pp. 135–147. DOI: 10.1016/j.cag.2010.11.005

Conference

- 1 Irina Paraschivoiu, Thomas Layer-Wagner, Klaus Neundlinger, Simone Rack, and **Markus Tatzgern**. “Where’s the Team Spirit? An Exploratory Study on Team Development Through Co-located Tablet-Based VR”. in: *Proceedings of ACM Human-Computer Interaction (PACMHCI), Special Issue ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)* 10.6 (2026), pp. 1–31. DOI: 10.1145/3817040
CORE A

- 2 Lucchas Ribeiro Skreinig, Denis Kalkofen, David Ritter, Peter Mohr, **Markus Tatzgern**, Dieter Schmalstieg, and Ana Stanescu. “Magnification Lenses for Small-Scale Augmented Reality Instructions”. In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. To appear. 2026

CORE A*

- 3 Helen Stefanidi[°], Jan-Hendrik Sünderkamp[°], **Markus Tatzgern**, Alina Itzlinger, and Alexander Meschtscherjakov. “Everyday ARactivities in-the-Wild: Insights from a Farmers Market Field Study”. In: *Proceedings of the Nordic Conference on Human-Computer Interaction (NordiCHI)*. ACM, 2026, pp. 1–11. DOI: 10.1145/3829807.3829863
- 4 Faith Young[°], Anna Lienbacher, Janne Mascha Beuthel, Erich Prem, Clemens Havas, and **Markus Tatzgern**. “Ethics and AI: Designing an Interactive Self-Assessment Tool for Ethical Implementation of AI-enabled Projects”. In: *Proceedings of the Nordic Conference on Human-Computer Interaction (NordiCHI)*. ACM, 2026, pp. 1–18. DOI: 10.1145/3829807.3829864
- 5 Jan-Hendrik Sünderkamp[°], Helen Stefanidi[°], Alexander Meschtscherjakov, and **Markus Tatzgern**. “From Lab to Sidewalk: Exploring AR Interaction Techniques Outdoors and in Motion using Fitts’ Law”. In: *Proceedings of ACM Symp. on Virtual Reality Software and Technology (VRST)*. 2025. DOI: 10.1145/3756884.3766006
- 6 Sebastian Hubenschmid, Johannes Zagermann, Robin Erb, Tiare Feuchtnner, Jens Grubert, **Markus Tatzgern**, Dieter Schmalstieg, and Harald Reiterer. “SpatialMouse: A Hybrid Pointing Device for Seamless Interaction Across 2D and 3D Spaces”. In: *Proceedings of ACM Symp. on Virtual Reality Software and Technology (VRST)*. 2025. DOI: 10.1145/3756884.3766047
- 7 Helen Stefanidi[°], Jan-Hendrik Sünderkamp[°], **Markus Tatzgern**, Alina Itzlinger, and Alexander Meschtscherjakov. “You’re making things AR-kward: Exploring Augmented Reality In-the-Wild”. In: *Proceedings of ACM Human-Computer Interaction (PACMHCI), Special Issue ACM International Conference on Mobile Human-Computer Interaction (MobileHCI) 9.5* (2025), pp. 1–24. DOI: 10.1145/3743740

Honorable Mention for Best Paper

- 8 Jan Hendrik Plümer[°], Kevin Yu, Ulrich Eck, Denis Kalkofen, Philipp Steininger, Nassir Navab, and **Markus Tatzgern**. “XR Prototyping of Mixed Reality Visualizations: Compensating Interaction Latency for a Medical Imaging Robot”. In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2024. DOI: 10.1109/ISMAR62088.2024.00014

CORE A*

Honorable Mention for Best Student Paper

- 9 Lucchas Ribeiro Skreinig, Peter Mohr, Blanca Berger, **Markus Tatzgern**, Dieter Schmalstieg, and Denis Kalkofen. “Immersive Authoring by Demonstration of Industrial Procedures”. In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2024, pp. 1293–1302. DOI: 10.1109/ISMAR62088.2024.00146

CORE A*

- 10 Helen Stefanidi[°], **Markus Tatzgern**, and Alexander Meschtscherjakov. “Augmented Reality on the Move: A Systematic Literature Review for Vulnerable Road Users”. In: *Proceedings of ACM Human-Computer Interaction (PACMHCI), Special Issue ACM International Conference on Mobile Human-Computer Interaction (MobileHCI) 8* (2024), pp. 1–30. DOI: 10.1145/3676490
- 11 Jan Hendrik Plümer[°] and **Markus Tatzgern**. “Towards a Framework for Validating XR Prototyping for Performance Evaluations of Simulated User Experiences”. In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2023, pp. 810–819. DOI: 10.1109/ISMAR59233.2023.00096

CORE A*

- 12 Gerlinde Emsenhuber[°], Tobias Langlotz, Denis Kalkofen, Jonathan Sutton, and **Markus Tatzgern**. “Eye-Perspective View Management for Optical See-Through Head-Mounted Displays”. In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. New York, NY, USA: Association for Computing Machinery, 2023. DOI: 10.1145/3544548.3581059
CORE A*
- 13 **Markus Tatzgern**, Michael Domhardt, Martin Wolf[†], Michael Cenger, Gerlinde Emsenhuber, Radomir Dinic, Nathalie Gerner, and Arnulf Hartl. “AirRes Mask: A Precise and Robust Virtual Reality Breathing Interface Utilizing Breathing Resistance as Output Modality”. In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. CHI ’22. New York, NY, USA: Association for Computing Machinery, 2022. DOI: 10.1145/3491102.3502090
CORE A*
Honorable Mention for Best Paper
- 14 **Markus Tatzgern** and Christoph Birgmann[†]. “Exploring Input Approximations for Control Panels in Virtual Reality”. In: *Proceedings of IEEE Virtual Reality (VR)*. 2021. DOI: 10.1109/VR50410.2021.00092
CORE A*
- 15 Masahiro Yamaguchi, Shohei Mori, Peter Mohr, **Markus Tatzgern**, Ana Stanescu, Hideo Saito, and Denis Kalkofen. “Video-Annotated Augmented Reality Assembly Tutorials”. In: *Proceedings of ACM Symposium on User Interface Software and Technology (UIST)*. 2020, pp. 1010–1022. DOI: 10.1145/3379337.3415819
CORE A*
- 16 Peter Mohr, **Markus Tatzgern**, Tobias Langlotz, Andreas Lang, Dieter Schmalstieg, and Denis Kalkofen. “TrackCap: Enabling Smartphones for 3D Interaction on Mobile Head-Mounted Displays”. In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. 2019, pp. 1–11. DOI: 10.1145/3290605.3300815
CORE A*
- 17 Peter Mohr, **Markus Tatzgern**, Jens Grubert, Dieter Schmalstieg, and Denis Kalkofen. “Adaptive User Perspective Rendering for Handheld Augmented Reality”. In: *IEEE Symposium on 3D User Interfaces (3DUI)*. 2017, pp. 176–181. DOI: 10.1109/3DUI.2017.7893336
Honorable Mention for Best Paper
- 18 Peter Mohr, David Mandl, **Markus Tatzgern**, Eduardo E. Veas, Dieter Schmalstieg, and Denis Kalkofen. “Retargeting Video Tutorials Showing Tools With Surface Contact to Augmented Reality”. In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. ACM, 2017, pp. 6547–6558. DOI: 10.1145/3025453.3025688
CORE A*
- 19 **Markus Tatzgern**, Valeria Orso, Denis Kalkofen, Giulio Jacucci, Luciano Gamberini, and Dieter Schmalstieg. “Adaptive information density for augmented reality displays”. In: *Proceedings of IEEE Virtual Reality (VR)*. 2016, pp. 83–92. DOI: 10.1109/VR.2016.7504691
CORE A*
- 20 **Markus Tatzgern**, Denis Kalkofen, Raphael Grasset, and Dieter Schmalstieg. “Hedgehog Labeling: View Management Techniques for External Labels in 3D Space”. In: *Proceedings of IEEE Virtual Reality (VR)*. 2014, pp. 1–6. DOI: 10.1109/VR.2014.6802046
CORE A*

- 21 **Markus Tatzgern**, Raphael Grasset, Denis Kalkofen, and Dieter Schmalstieg. “Transitional Augmented Reality Navigation for Live Captured Scenes”. In: *Proceedings of IEEE Virtual Reality (VR)*. 2014, pp. 21–26. DOI: 10.1109/VR.2014.6802045
CORE A*
- 22 **Markus Tatzgern**, Denis Kalkofen, and Dieter Schmalstieg. “Dynamic compact visualizations for augmented reality”. In: *Proceedings of IEEE Virtual Reality (VR)*. 2013. DOI: 10.1109/VR.2013.6549347
CORE A*
- 23 **Markus Tatzgern**, Raphael Grasset, Eduardo Veas, Denis Kalkofen, Hartmut Seichter, and Dieter Schmalstieg. “Exploring Distant Objects with Augmented Reality”. In: *Joint Virtual Reality Conference of EGVE - EuroVR*. ed. by Betty Mohler, Bruno Raffin, Hideo Saito, and Oliver Staadt. 2013, pp. 1–11. DOI: 10.2312/EGVE.JVRC13.049-056
- 24 Raphael Grasset, Tobias Langlotz, Denis Kalkofen, **Markus Tatzgern**, and Dieter Schmalstieg. “Image-Driven View Management for Augmented Reality Browsers”. In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2012, pp. 177–186. DOI: 10.1109/ISMAR.2012.6402555
CORE A*
- 25 **Markus Tatzgern**, Denis Kalkofen, and Dieter Schmalstieg. “Compact Explosion Diagrams”. In: *Proceedings of the International Symposium on Non-Photorealistic Animation and Rendering*. 2010, pp. 17–26. DOI: 10.1145/1809939.1809942
- 26 Denis Kalkofen, **Markus Tatzgern**, and Dieter Schmalstieg. “Explosion Diagrams in Augmented Reality”. In: *Proceedings of IEEE Virtual Reality (VR)*. 2009, pp. 71–78. DOI: 10.1109/VR.2009.4811001
CORE A*

Posters, Workshops, Demos and Other Publications

- 1 Lucchas Ribeiro Skreinig, Ana Stanescu, Peter Mohr, Christoph Ebner, **Markus Tatzgern**, Dieter Schmalstieg, and Denis Kalkofen. “Augmented Reality Action Summaries for Situated Instructions of Repetitive Tasks”. In: *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. Poster, to appear. 2026
- 2 Faith Young[◊], **Markus Tatzgern**, Alexander Meschtscherjakov, and Jan David Smeddinck. “Conceptualising an Initial Design Space for Guidance in Digital Physical Activity Support”. In: *Adjunct Proceedings of the Nordic Conference on Human-Computer Interaction (NordiCHI Adjunct)*. Poster. ACM, 2026, pp. 1–9. DOI: 10.1145/3821402.3830134
- 3 Helen Stefanidi[◊], Vincent van Rheden, Linjia He, Michelle Adiwangsa, Jan-Hendrik Sünderkamp[◊], Alina Itzlinger, Andrii Matviienko, Julie R. Williamson, Ian Oakley, **Markus Tatzgern**, and Alexander Meschtscherjakov. “Next Steps for Augmented Reality On-the-Move: Challenges and Opportunities”. In: *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA)*. CHI EA ’26. ACM, 2026. DOI: 10.1145/3772363.3778775
- 4 Sebastian Hubenschmid, Johannes Zagermann, Niklas Elmqvist, Tiare Feuchtner, Jens Grubert, **Markus Tatzgern**, Dieter Schmalstieg, and Harald Reiterer. “Revisiting Hybrid Input Devices for Immersive Analytics”. In: *IEEE Conference on Human Factors in Immersive Analytics (HFIA), IEEE VIS Workshop*. 2025, pp. 29–33. DOI: 10.1109/HFIA68651.2025.00011

- 5 Gerlinde Emsenhuber[°], Michael Smirnov, Tobias Langlotz, Denis Kalkofen, Stefanie Zollmann, and **Markus Tatzgern**. “Mobile Eye-perspective Rendering: Alignment of Image-based Vision Augmentations for Optical See-through Head-mounted Displays”. In: *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. 2025, pp. 917–918. DOI: 10.1109/ISMAR-Adjunct68609.2025.00250
- 6 Helen Stefanidi[°], Jan-Hendrik Sünderkamp[°], Alina Itzlinger, **Markus Tatzgern**, and Alexander Meschtscherjakov. “Exploring AR In-the-Wild: An Autoethnographic Study at a Christmas Market”. In: *Companion Publication of the 2025 ACM Designing Interactive Systems Conference (DIS)*. 2025, p. 7. DOI: 10.1145/3715668.3736368
- 7 Helen Stefanidi[°], **Markus Tatzgern**, Alina Itzlinger, and Alexander Meschtscherjakov. “Bringing Adaptive On-the-Move AR to Outdoor Environments”. In: *ACM Conference on Human Factors in Computing Systems (CHI), Workshop Artificial Intelligence and Human Interfaces*. 2025
- 8 **Markus Tatzgern**, David Fasching, Vivian Gómez, Sebastián Calvache, and Pablo Figueroa. “Passive Haptic Feedback for Control Elements in Virtual Reality”. In: *IEEE Virtual Reality and 3D User Interfaces - Abstracts and Workshops (VRW)*. 2024, pp. 749–750. DOI: 10.1109/VRW62533.2024.00171
- 9 Gerlinde Emsenhuber[°], Michael Domhardt, Tobias Langlotz, Denis Kalkofen, and **Markus Tatzgern**. “Towards Eye-Perspective Rendering for Optical See-Through Head-Mounted Displays”. In: *IEEE Virtual Reality and 3D User Interfaces - Abstracts and Workshops (VRW)*. 2022, pp. 640–641. DOI: 10.1109/VRW55335.2022.00171
- 10 Denis Kalkofen, Shohei Mori, and **Markus Tatzgern**. “Visualization and Graphics in Mixed Reality”. In: *Eurographics - Tutorials*. The Eurographics Association, 2021. DOI: 10.2312/egt.20211036
- 11 **Markus Tatzgern**, Denis Kalkofen, Raphael Grasset, and Dieter Schmalstieg. “Embedded Virtual Views for Augmented Reality Navigation”. In: *Workshop on Visualization in Mixed Reality Environments, International Symposium on Mixed and Augmented Reality (ISMAR)*. 2011. DOI: 10.13140/2.1.4398.1769

Patents

- 2025 Undisclosed patent pending (Qualcomm). Markus Tatzgern, Gerlinde Emsenhuber.
- 2014 Hierarchical clustering for view management augmented reality. Markus Tatzgern, Denis Kalkofen, Dieter Schmalstieg, Raphael Grasset. US Patent, Publication number 14/640,981, 2014
- 2013 Image-Driven View Management for Annotations. Raphael Grasset, Markus Tatzgern, Tobias Langlotz, Denis Kalkofen and Dieter Schmalstieg. WO Patent, Publication number 2013176830, November 2013
- 2012 Efficient Information Presentation for Augmented Reality. Markus Tatzgern, Denis Kalkofen and Dieter Schmalstieg. WO Patent, Publication number 2012033768, March 2012

Supervision

Official supervision and co-supervision of five PhD students, four of which are located at Salzburg University of Applied Sciences.

- ongoing Supervision of PhD thesis by Gerlinde Emsenhuber,
Eye-perspective Rendering for Optical See-through Head-mounted Displays
Located at Salzburg University of Applied Sciences, PhD programme at Graz University of Technology with Prof. Denis Kalkofen
- ongoing Supervision of PhD thesis by Jan Hendrik Plümer,
Validation of XR Prototyping for Human-robot Interaction
Located at Salzburg University of Applied Sciences, PhD programme at Graz University of Technology with Prof. Denis Kalkofen
- ongoing Supervision of PhD thesis by Jan-Hendrik Sünderkamp,
Effective XR Interaction while Being on the Move
Located at Salzburg University of Applied Sciences, PhD programme at University of Salzburg with Prof. Alexander Meschtscherjakov
- ongoing Co-supervision of PhD thesis by Faith Young,
Ethical AI-based Health Interventions
Located at Salzburg University of Applied Sciences, PhD programme at University of Salzburg with Prof. Alexander Meschtscherjakov
- ongoing Co-supervision of PhD thesis by Helen (Eleni) Stefanidi,
XR User Experiences in Outdoor Environments
PhD programme at University of Salzburg with Prof. Alexander Meschtscherjakov

Regular supervision of Bachelor's theses (53 completed) and Master's theses (24 completed).

Teaching

Salzburg University of Applied Sciences, Bachelor Level

- 2015–2026 Game Development 1 (Lecture and Lab)
- 2016–2020 Game Development 2 (Lecture and Lab)
- 2016–2021 Computer Graphics (Lecture and Lab)
- 2016 Advanced Topics of Game Development and Augmented Reality (Lecture and Lab)
- 2016–2019 Study & Presentation Skills (Scientific Writing)
- 2020–2026 Seminar Paper Computer Science (Scientific Writing)
- 2016–2026 Bachelor Thesis Seminar
- 2024–2026 Augmented & Virtual Reality (Lecture and Lab)

Salzburg University of Applied Sciences, Master Level

- 2017–2026 Mixed Reality Technologies (Lecture and Lab)
- 2016 Mixed Reality Interaction (Lecture and Lab)
- 2017–2024 Real-Time Visualization (Lecture and Lab)
- 2020, 2025 Interaction Approaches & Technologies (Lecture and Lab)
- 2023 Applied Games (Lecture and Lab)
- 2024 Guest Lecture Game & Simulation Engineering (Lecture and Lab)
- 2019–2023 Research Seminar
- 2024–2026 Master Thesis Seminar 1
- 2024–2026 Research Method and Study Design
- 2016–2026 Master Thesis Seminar 2

Professional Development

- 10/2025 Gender and Diversity Training
1 half-day course
- 04/2023–06/2023 Leadership Training
Seminar series covering 2.5 full-day modules.
- 09/2016 Higher Education Didactics: Instructional Design
1 half-day module
- 10/2015 Higher Education Didactics: Definition and Assessment of Learning Outcomes
1 full-day module
- 10/2015 Higher Education Didactics: Supervision of Thesis Work
1 full-day module
- 10/2015 Higher Education Didactics: Reflective Learning. Stimulating Conscious Learning Processes
1 half-day module
- 09/2015 Higher Education Didactics: Didactics of an Online Course
1 full-day module
- 09/2015 Higher Education Didactics: Exams and Skill Acquisition
1 full-day module
- 12/2014 Protecting Intellectual Property: Patent Applications and Exploitation
1 full-day module
- 11/2014 EU Project Acquisition and Management
1 full-day module

Awards and Honours

- 2025 Honorable Mention for Best Paper at the ACM International Conference on Mobile Human-Computer Interaction (MobileHCI), Paper: You're making things AR-kward: Exploring Augmented Reality In-the-Wild
- 2024 Honorable Mention for Best Student Paper at the IEEE Symposium for Mixed and Augmented Reality (ISMAR) (**CORE A***), Paper: XR Prototyping of Mixed Reality Visualizations: Compensating Interaction Latency for a Medical Imaging Robot
- 2022 Honorable Mention for Best Paper at the ACM CHI Conference on Human Factors in Computing Systems (CHI) (**CORE A***), Paper: AirRes Mask: A Precise and Robust Virtual Reality Breathing Interface Utilizing Breathing Resistance as Output Modality
- 2017 Honorable Mention for Best Paper at the IEEE Symposium on 3D User Interfaces (3DUI), Paper: Adaptive User Perspective Rendering for Handheld Augmented Reality
- 2011, 2013, 2015 Distinct Inventor at Graz University of Technology

Editorial and Conference Leadership

Chair Duties

- 2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Daejeon, South Korea, Science and Technology Papers Chair for Conference and Journal Track (transition to single track ISMAR)

- 2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Greater Seattle Area, Science and Technology Program Chair for Conference and Journal Track
- 2024, 2025 ACM Conference on Human Factors in Computing Systems (CHI), Associate Chair, Subcommittee User Experience and Usability
- 2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Sydney, Poster Chair

Editorial Duties

- since 2026 Associate Editor of IEEE Transactions on Visualization and Computer Graphics
- since 2024 Editorial Board of Springer Journal Virtual Reality
- since 2021 Editorial Board of Journal Virtual Reality & Intelligent Hardware
- since 2020 Editorial Board of Augmented Reality: Review Editor for Frontiers in Virtual Reality

Service

Committee Duties

- 2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- 2026 European Association for eXtended Reality (EuroXR) International Conference
- 2026 IEEE Visualization and Graphics Technical Community (VGTC) Virtual Reality Best Dissertation Award Committee Member
- 2023, 2025, 2027 IEEE Virtual Reality and 3D User Interfaces (VR), conference and journal track (IEEE Transactions on Visualization and Computer Graphics)
- 2022 IEEE International Conference on Artificial Intelligence & Virtual Reality (AIVR)
- 2022 IEEE Virtual Reality and 3D User Interfaces (VR), journal track (IEEE Transactions on Visualization and Computer Graphics)
- 2021 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), journal track (IEEE Transactions on Visualization and Computer Graphics)
- 2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), conference track
- 2016, 2019–2022 IEEE Virtual Reality and 3D User Interfaces (VR)
- 2019 ACM Symposium on Virtual Reality Software and Technology (VRST)
- 2018 FH Forschungsforum, Chair: Digitized - Transformation durch Augmented und Virtual Reality

Other Roles

- 2026 AVRIL 2026, Committee Member for Competition XR Learning for German Informatics Society (GI)
- since 10/2025 Chapter Officer (Treasurer) of Austrian ACM Special Interest Group on Computer-Human Interaction (SIGCHI) Chapter
- 2025 Founding Member of Forum New Media Austria (FNMA) Special Interest Group Immersive Education
- 2025, 2026 EdTech Austria Fellowship Program - Mentor
- since 04/2025 Member of advisory board in FFG funded research project GreenTouch - XR in Ergotherapy, Austria

- since 10/2023 Scientific Collaboration and Consulting in Digital Health Technologies, Ludwig Boltzmann Institute - Digital Health and Prevention, Austria
- since 2023 Founding Member of Austrian ACM Special Interest Group on Computer-Human Interaction (SIGCHI) Chapter
- 2023 Jury Member for the Content Vienna Award funded by the Vienna Business Agency, Austria
- 2022 Expert Evaluator for Accreditation of Study Program for University of Applied Management, Germany
- 2020, 2021, 2026 Doctoral Consortium Mentor at IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- 2017, 2018 Expert Evaluator for Austrian Research Promotion Agency (FFG)
- Reviewing**
- since 2009 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- since 2011 IEEE Virtual Reality (VR)
- since 2011 Computers & Graphics (C&G)
- since 2013 ACM Conference on Human Factors in Computing Systems (CHI)
- since 2017 IEEE Transactions on Visualization and Computer Graphics
- since 2016 ACM Symposium on Virtual Reality Software and Technology (VRST)
- 2025 Springer Journal Virtual Reality
- 2023 International Conference on Human-Computer Interaction (INTERACT)
- 2022, 2023, 2026 Doctoral Consortium Reviewer at IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- 2020 ACM Symposium on User Interface Software and Technology (UIST)
- 2022 IEEE Computer Graphics and Applications (CG&A)
- 2012 Eurographics
- 2012 ACM International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI)
- 2015 ACM Symposium on Spatial User Interaction (SUI)
- 2015 International Conference on Mobile and Ubiquitous Multimedia (MUM)
- 2016 Journal of Virtual Reality and Broadcasting (JVRB)
- 2020 ACM Interaction Design and Children (IDC)
- Reviewing Recognitions**
- 2025, 2026 Special Recognition for Outstanding Reviews for ACM CHI Conference on Human Factors in Computing Systems (CHI), Papers Track
- 2024 Reviewer Certificate for reviewing efforts from IEEE Transactions on Visualization and Computer Graphics (TVCG)
- 2020 Special Recognition for Outstanding Reviews for ACM Symposium on User Interface Software and Technology (UIST), Papers Track
- 2020 Special Recognition for Outstanding Reviews for IEEE Symposium for Mixed and Augmented Reality, Papers Track
- 2019, 2021, 2022 Recognition for 'highly useful' reviews for IEEE Symposium for Mixed and Augmented Reality, Journal Papers Track

Invited Talks and Presentations

- 2026 Invited talk, “Everyday XR - Moving Out of the Lab”, Wintergraph 2026, 19th Visual Computing Winter Workshop, Austria
- 2025 Panel discussion, “Immersion x Education for Youth”, culttech summit, Vienna
- 2024 Invited talk, “EdTech - Augmented Learning Lab”, EdTech Summit, Salzburg
- 2022 Invited talk and panel discussion, “Trends and Potentials of EdTech”, EdTech Austria Roadshow
- 2021 Invited talk, “Supporting Primary School Education with Technology”, EdTech Austria - Round Table
- 2021 Tutorial, “Visualization and Graphics in Mixed Reality”, Eurographics 2021
- 2018 Invited talk, “Digitalization in Architecture and Construction - Use Cases of Virtual and Augmented Reality”, Brennpunkt Alpines Bauen, Austria
- 2018 Invited talk, “Virtual and Augmented Reality - Technological Maturity and Applications in Tourism”, Brennpunkt eTourism, Austria
- 2018 Invited talk, “State-of-the-Art of Virtual and Augmented Reality Technology and Applications”, Austrian Armed Forces, Austria
- 2018 Invited talk, “Virtual Reality in E-commerce”, Marketing Club Salzburg, Austria
- 2018 Invited talk, “Augmented and Virtual Reality for Medical Education and Health Care”, 1. Salzburger Forum für Medizin und Digitalisierung, Paracelsus Medical University, Salzburg, Austria
- 2017 Invited talk, “Augmented and Virtual Reality: A building block of industry 4.0”, Alles Holz 4.0 oder was, Holzcluster, Salzburg, Austria
- 2017 Invited talk, “Virtual and Augmented Reality in Education”, eEducation Praxistagung, Pädagogische Hochschule Salzburg, Salzburg, Austria
- 2015 Invited talk, “Augmented Reality Technology and Applications”, Center for Human-Computer Interaction, University of Salzburg, Salzburg, Austria
- 2015 Invited talk, “Next Generation Augmented Reality Games”, Extended Games Day, Hochschule der Medien, Stuttgart, Germany

Public Engagement (Selected Media)

- 2025 “Artificial, intelligent and educational?” / “Künstlich, intelligent – und bildend”, Salzburger Nachrichten (print, online), 16 December, Austria
- 2025 “Education in Salzburg making headway for using AI for teaching” / “Salzburger Bildung macht sich fit für KI in der Lehre”, Salzburger Nachrichten (print, online), 10 November, Austria
- 2024 “FH Salzburg presents new research strategy”, MyScience (online), 6 June
- 2023 “Saving time and money utilizing virtual prototyping” / “Mit virtuellen Prototypen Zeit und Kosten sparen”, Salzburger Nachrichten (print), 20 October, Austria
- 2022 “Introducing ‘AirRes Mask’ that can introduce the concept of ‘breathing’ into VR”, Gigazine (online), 16 May, Japan
- 2022 “This mask makes breathing in virtual reality more realistic”, edition.cnn.com / CNN Business (online), 15 May

- 2022 "Virtual reality mask adds realism by making it harder to breath", NewScientist, newscientist.com, 11 May
- 2021 "The potential of Augmented Reality" / "Wie viel Potenzial steckt in Augmented Reality", doIT digitale Karriere-Perspektiven, Der Standard (print), 27 November, Austria
- 2021 "Learning with digital support" / "Lernen gelingt mit digitalen Helfern", Salzburger Nachrichten (print), 24. April, Austria
- 2021 "Fannys Friday: Game Science", ORF / Austrian Broadcasting Corporation (television) - tvthek.orf.at, 8 January, Austria
- 2020 "Teaching Chemistry in a Virtual Laboratory"/"Chemie vermitteln lernen im virtuellen Labor", Die Presse (print) - diepresse.com, 12 December, Austria
- 2019 "Augmented Reality in the Work Place"/ TV report "Austrian Broadcasting Corporation" ("Österreichischer Rundfunk"), 4 March, Austria
- 2018 "New Carnegie Museum app shows threat to wildflower diversity in the woods", Pittsburgh Post-Gazette / post-gazette.com, 26 December, USA
- 2018 "Augmented Reality App Shows How Deer Overpopulation Affects Western PA's Forests", Pittsburgh's NPR News Station / wesa.fm, 19 December, USA